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Supplementary Materials for

The fiscal impact of biodiversity loss and a pathway for conservation finance

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1. Materials and methods.

Our analyses require data on White Nose Syndrome (WNS), county-level numbers of WNS-susceptible bat species, assessed frequencies of bat occurrence, county rural-urban designations, county bond spreads, county revenue and tax assessments, and additional conditioning variables. We describe these data in section 1.1 and our empirical approach in section 1.2. Section 1.3 discusses the basics of bond pricing. Code to reproduce the analysis and links to data can be found at `https://github.com/AnnaNakhmurina/wns_and_conservation_finance/tree/main`.

1.1. Data.

1.1.1. White nose syndrome.

We obtain data on the location and timing of WNS detection from the White Nose Syndrome Response Team, a centralized repository of WNS-related information that sources its data from state and federal authorities.[1]

The White Nose Syndrome Response Team classifies counties into 5 groups: "*Pseudogymnoascus destructans* (Pd) Positive," "WNS Positive," "WNS Suspect," "Pd Presumed,"

[1] `whitenosesyndrome.org`

and "[null]", with the last category indicating that no WNS or Pd was found in a county.[2]

Prior research documented a two-year lag between the actual emergence of WNS in a county and its official detection (Frank, 2024). Following this approach, we classify a county as "treated" beginning two years before it was designated as "WNS positive" or "WNS suspect" by the White Nose Syndrome Response Team. Counties where only the underlying fungus, *Pseudogymnoascus destructans*, has been detected remain classified as untreated in all years. Our treatment classification, therefore, is county-year-specific. For brevity, we use the term "treated" counties, though this actually refers to county-year observations where WNS is present. In our main research design (described in section 1.2), unaffected counties, counties already experiencing WNS, and counties not yet affected by WNS serve as control groups. In section 2.4, we test for robustness against known issues, such as the inclusion of not-yet-affected counties in the control group.

Following Frank (2024), we exclude counties that neighbor WNS-positive counties, but never experience WNS. This mitigates concerns related to spillover effects.

The distribution of WNS across the lower 48 states used in our analysis is shown in Figure S1.

[2]The White Nose Syndrome Response team defines *WNS Positive* when "characteristic histologic lesions of WNS are present on an individual bat and the bat is positive for *Pseudogymnoascus destructans* (Pd) either by qPCR2 or by fungal culture. *WNS Suspect* is defined when one of the following criteria are met (a) Characteristic histologic lesions of WNS1 are present on an individual bat but Pd is not detected, the test result for Pd is inconclusive (either by qPCR2 or by fungal culture), or testing for Pd is not performed; (b) One or more field signs are observed in a bat colony and Pd is detected (either by qPCR2, fungal culture, or a tapelift performed directly on visible fungal growth on bat skin) on at least one individual of the same species and histopathology is negative or is not performed; (c) Multiple field signs are observed among species known to be susceptible to WNS and are within the currently recognized range of WNS but no samples are collected for diagnostic evaluation; or (d) Individual bats that are part of a confirmed WNS morbidity or mortality event are submitted to, but not tested by, a diagnostician. This criterion is for instances in which multiple samples from the same site are submitted, but only a subset of those samples is tested. The untested samples may be classified as suspect for WNS if the subset of tested samples is positive for WNS and consists of the same species as the untested samples."

1.1.2. *Bat species richness.*

In addition to WNS data, we use data on bat species richness and bat detection probabilities by species. These data are primarily used in robustness checks and to understand herogeneous effects. Bat species richness data were provided by Frank (2024) and used to create an indicator variable equal to one if a county has an above-median number of bat species susceptible to WNS.[3]

In our robustness checks we use the North American Bat Monitoring Program (NABat) data set.[4] NABat data approximate the probability of observing 12 bat species over space and time (Udell et al., 2022). However, data only exist from 2016 to 2019 for most species in this data set, limiting its usefulness in our study. We focus on the occurrence probabilities of the big brown bat (*Eptesicus fuscus*), one of the species most affected by WNS, common generally, and particularly prevalent over agricultural fields (Fill et al., 2022). We aggregate sub-county detection probabilities of the big brown bat from NABat to the county level using averages.

1.1.3. *Rural county classification.*

We categorize a county as rural if, as of 2003, the U.S. Department of Agriculture classified it as (1) a non-metro county completely rural or less than 2,500 urban population, both adjacent and not adjacent to a metro area, (2) a non-metro county with urban population of 2,500 to 19,999, both adjacent and not adjacent to a metro area, or (3) a non-metro county with urban population of 20,000 or more, both adjacent and not adjacent to a metro area.

[3]We thank Eyal Frank for sharing these data, calculated from the USGS Gap Analysis Project.
[4]https://www.nabatmonitoring.org/.

We choose the 2003 classification because it is closest to the beginning of our sample. We explore alternative definitions of a rural county in our robustness checks.

1.1.4. Bond data and bond variable definitions.

Data on municipal bond primary offerings come from the Mergent Municipal database (Mergent).[5] This database contains the bond characteristics that we use as control variables in our regressions, including bond yields at the time of offering, bond size, time to maturity, credit rating, bond type (e.g., general obligation or revenue bond), and information about whether the bond is callable, insured, and bank qualified. These are all standard controls in the municipal bond literature (e.g., (Schwert, 2017; Painter, 2020)). We define these variables in the next few paragraphs. For a general overview of the municipal bond market, see (Cestau et al., 2019).

Bond size is the principal amount borrowed. Time to maturity is the length of time until the principal is repaid. General obligation bonds are backed by the full taxing power of the county, while revenue bonds are backed by a pre-specified revenue stream. Callable bonds are bonds that can be paid off or redeemed by the issuer (i.e., county) prior to the bond's maturity date. Insured bonds are backed by an insurance company that guarantees all scheduled interest and principal payments will be made to bondholders, even if the original issuer defaults. Bank-qualified municipal bonds receive preferential tax treatment when commercial banks purchase them.

A credit rating is the underlying rating assigned by one of the credit rating agencies (Moody's, S&P, or Fitch) at the time of issuance. To ensure comparability across agencies,

[5] For a description of these data, see `https://bv.mergent.com/view/scripts/common/index.php`.

we harmonize their letter ratings into 22 categories. For bonds rated by multiple agencies, we calculate the rounded average of these numeric ratings to measure the underlying credit quality. We include fixed effects for these aggregate credit rating measures in our regressions.

We restrict these data to county-issued bonds by manually identifying the issuer-specific identifiers (six-digit CUSIPs) associated with county issuers.[6] We exclude variable-rate and taxable bonds to ensure a consistent comparison across our sample (Green et al., 2010).

We define municipal bond spread as the difference between the offering yield and the maturity-matched AAA-rated Municipal Market Advisors yield curve (MMA, downloaded from Bloomberg). For example, we match a 10-year maturity bond with the 10-year rate provided by MMA. For maturities where the MMA data are unavailable, we perform a linear interpolation. To account for different state-level tax regimes, spreads are tax-adjusted following Schwert (2017). The resulting offering yield spreads can be interpreted as interest rates on municipal debt that account for the risk-free rate and state-specific capital tax differences, making them comparable across different times and states. Because the MMA curve starts in 2001 and the Mergent data end in 2022, our final bond sample spans 2001 to 2022.

Summary statistics for the bond data are in Table S1. Panel A reports summary statistics, showing that the average spread for rural counties issuing bonds is 42.03 basis points.[7] This means, for example, that a rural county paying the average spread would have a yield of 2.92% when the corresponding AAA MMA yield is 2.50%. 74% of our sample

[6]CUSIP (Committee on Uniform Security Identification Procedures) is an alphanumeric code system that is used to uniquely identify financial securities. Nine-digit CUSIPs identify securities, while the first six digits of CUSIPs identify issuers.

[7]A basis point is one hundredth of a percentage point.

bonds are bank qualified, 73% are general obligation, 46% are callable, and 36% are insured. The average time to maturity is 8.85 years and ranges from 2 to 17 years for the 10th and 90th percentiles, respectively. For a single bond, the average amount borrowed is $0.55 million. Municipalities often borrow in bond issues that contain multiple bonds. The average offering size of an issue consisting of multiple bonds is $8.34 million (see Panel C of Table S1). Panel B shows the correlations among the variables. Panel C provides summary statistics used in the hypothetical investment examples.

1.1.5. Secondary bond market data.

In our robustness analyses, we use secondary market transaction data. We obtain these data from the Municipal Securities Rulemaking Board (MSRB) and apply standard data cleaning procedures following the municipal bond literature (Green et al., 2010; Schwert, 2017). To eliminate data errors, we remove transactions with missing coupon or maturity information and exclude trades recorded on weekends or holidays. We also exclude bonds with coupons exceeding 20% and maturities longer than 100 years. Additionally, we drop transactions with dollar prices above $150 for bonds with less than one year to maturity. We restrict our sample to bonds with at least one year to maturity and require a minimum of 10 trades over each bond's lifetime. Finally, we construct monthly trade-weighted yields following Cornaggia et al. (2018) and calculate bond spreads as we did for primary bond market spreads.

1.1.6. *County revenue data.*

County-level government revenue and socio-demographic data come from The Government Finance Database, United States, 2001-2022 (see Table S2 for the corresponding summary statistics). We use this time period to maintain consistency with our bond offering data described above.[8] The sample of revenue data is nearly complete for the continental United States (Figure S1), with very few counties having missing data over the full time period. To connect the change in the ecological system (bats) to agricultural land markets, we focus on property taxes and own-source revenue (revenues raised from local sources, excluding state and federal transfers). To compare across counties of different sizes, we normalize revenue variables by county population and focus on per capita impacts (e.g., Dagostino and Nakhmurina (2023)).

Our analyses focus on rural counties, which rely heavily on agricultural property tax revenue (see separate summary statistics for rural and urban counties in Table S3, Panel A). Both the mean and median percentage of property tax collected from agricultural properties (calculated from CoreLogic *Tax* data) are substantially higher for rural counties. We also present summary statistics comparing rural and urban counties for bond variables (Panel B) and revenue variables (Panel C) in Table S3.

1.1.7. *Weather and sociodemographic data.*

To limit the influence of potential confounders, we control for the natural logarithm of the human population, as humans can inadvertently transmit WNS between bat populations through contaminated boots and equipment.

[8] Pierson et al. (2015) describes the construction of this dataset from the US Census.

We also control for minimum quarterly precipitation levels and temperatures. Temperature and precipitation impact agricultural output, while cold temperatures, particularly in winter, can affect WNS transmission. These data come from the Parameter-elevation Regressions on Independent Slopes Model (PRISM) data.[9]

Summary statistics for population and weather variables are provided in Tables S2 and S3.

In some specifications, we control for the number of neighboring counties with WNS in any year. This accounts for any spillover effects from neighboring counties that could affect outcomes when treatment occurs

1.1.8. Tax assessment data.

In our robustness analyses, we use property tax assessment data from CoreLogic *Tax*. This database tracks property tax assessments throughout the US by compiling county records. The data are a property-level panel that contains a unified parcel identifier, land use code, assessment date, assessment amount, and the amount of taxes paid in each tax year. We restrict our data to land use codes from 400 to 600, which are associated with agricultural activity.

We also use these data to infer the relative frequency of property tax assessments.

1.2. Empirical approach.

Our linear regression model (discussed in the main text) estimates the impact of WNS by comparing county outcomes before and after WNS detection (treated counties) relative

[9]These data can be accessed at`http://www.prism.oregonstate.edu/`.

to counties that never experienced WNS, counties that have not yet detected the disease, and those where WNS has been detected in earlier time periods (control counties). This staggered difference-in-differences approach allows us to isolate the effect of WNS by tracking how county finances change when the disease arrives. Our primary specification is a two-way fixed effects model, where we control for time-invariant characteristics of counties and common shocks that vary over time. We estimate equation SI-1:

$$y_{ct} = \beta_0 Post_{ct} + \beta_1 X_{ct} + \delta_t + \alpha_c + \varepsilon_{ct}, \tag{SI-1}$$

where y_{ct} is the county outcome of interest (bond spread, property tax per capita, or own revenue per capita) for county c in year t. $Post_{ct}$ is an indicator equal to 1 if a county has WNS in or before year t. X_{ct} includes controls for factors that could correlate with WNS arrival in a county and affect government finances. δ_t and α_c are year and county fixed effects, respectively. To limit the influence of outliers, we winsorize all continuous variables at the 1% level. Standard errors are clustered at the county level for bond and revenue analyses.

We begin by estimating the average impact of WNS on municipal bond spreads and property tax revenues per capita. We discuss the results of these regressions in the main text. Here, we also estimate the impact of WNS on own revenues per capita and test the robustness of our coefficient estimates to alternative datasets and model specifications (details in section 2.4 below).

Then, in the main text, we examine variation in treatment effects based on two county characteristics: (1) whether a county has an above-median number of bat species susceptible

to WNS as defined in Frank (2024), and (2) whether a county performs relatively frequent updates of assessed property values. In section 2.4 of this SM, we expand on the role of county characteristics. Specifically, we focus on the role of big brown bats in a county, dividing counties with high probability of big brown bat detection from those with low probability of detection.

We expect stronger impacts in counties with a greater stock of natural assets at risk, which we proxy with an indicator of having an above-median number of WNS-susceptible bat species. Counties with more vulnerable species or those dependent on a common vulnerable species (i.e., the big brown bat) likely experience more severe damage to their overall bat populations. Such counties may experience greater reductions in farm profitability, subsequently leading to more pronounced declines in property tax revenue and larger bond spread reactions.

We also expect stronger impacts in counties that update their land assessments more frequently because these counties incorporate changes in agricultural property values into their tax base more quickly. When bat populations decline, the resulting reduction in ecosystem services quickly translates to lower property tax revenues in counties with frequent reassessments, while counties with infrequent reassessment cycles may experience larger delays.

For the difference-in-differences approach to identify the marginal effect of the treatment, a maintained assumption is that the treated groups would have followed similar (parallel) trends to the untreated groups if it were not for the treatment. We cannot directly test this identifying assumption. However, we can examine whether treatment and control counties exhibited similar trends before WNS arrival. To do this, we modify equation SI-1 to include

both leads and lags of WNS exposure:

$$y_{ct} = \sum_k \gamma_k \mu_{ct} + \beta_1 X_{ct} + \delta_t + \alpha_c + \varepsilon_{ct}, \tag{SI-2}$$

where μ_{ct} are binary indicators equal to one when the time since initial WNS detection equals k and zero otherwise. Here, we consider $k \in \{-6, ..., 6\}$. We omit the indicator for the first year with WNS. We also present alternative estimates of treatment effects for the treated subpopulation to address econometric challenges in the presence of staggered treatment effects (Cunninham, 2021) in section 2.4 of this SM.

1.3. Bond pricing.

Here, we describe the standard bond pricing procedure. The price of a bond is the expected present value of interest payments made to bondholders over the life of the bond plus the expected present value of the principal (also called the maturity value) paid when the bond matures. Typically, interest payments are called coupon payments, and the principal is paid in full at maturity. To fix ideas, let the time to maturity be T periods and the maturity amount be M. The bond also has a coupon rate that reflects the cost of borrowing, equal to c. The coupon rate determines the coupon amount paid every time period, equal to $C = c \times M$. Bonds typically pay coupons semi-annually, meaning that T equals twice the number of years until maturity.

The bond price, P, depends on the bond-specific risk-adjusted discount rate, r, which reflects bond-specific default risk. Since coupons are paid semiannually, r is half the annual discount rate. The value r is conventionally referred to as "yield." The price of a bond is:

$$P = C \times \frac{1-(1+r)^{-T}}{r} + \frac{M}{(1+r)^T}. \tag{SI-3}$$

The first term in equation SI-3 describes the present value of all coupon payments ($C \times \sum_{t=1}^{T} \frac{1}{(1+r)^t}$) and the second term is the present value of receiving the principal at maturity, M, after T periods. The price, P, is decreasing in yield, r. If $c = r$ then the coupon rate, which is the interest charged, aligns with the risk-adjusted discount rate and $P = M$. When $c > r$ the bond sells at a premium and $P > M$. This can happen because the coupon payments exceed the risk-adjusted opportunity cost of capital. In contrast, when $c < r$ the bond is discounted. The discount reflects the fact that the coupon payments do not cover the risk-adjusted opportunity cost of capital.

2. Supplementary text.

The main results for the bond and property tax analyses are presented in Figures 1 and 2 of the main text, respectively. We tabulate the underlying regressions along with closely related variations of the main specifications in Tables S4, S5, and S6. We describe these results in sections 2.1 and 2.2.

We describe the investment case study in section 2.3, and discuss robustness across various dimensions in section 2.4.

2.1. *Main estimation results.*

In Table S4, Panel A, we present the average effect of WNS on municipal bond spreads. Column 1 reports the results of our simplest specification, including only county and year fixed effects. County fixed effects isolate the impact of WNS on bond spreads after subtracting any time-invariant county characteristics that may affect its borrowing costs. Year fixed effects remove any time-specific factors that could have influenced bond spreads in a given year. Using this specification, we find that WNS-affected counties face 10.32 basis points higher primary offering spreads than control counties.

As we move from column 1 to column 6, we introduce additional control variables that account for bond characteristics, including the number of neighboring counties affected by WNS, the natural logarithm of bond size, the natural logarithm of time to maturity, as well as indicators for whether the bond is bank-qualified, general obligation, callable, and insured. We also include weather controls (quartiles of minimum temperature and precipitation, not tabulated to keep the tables concise), bond rating fixed effects, and the natural logarithm of human population. The gap in offering spreads between WNS-affected and unaffected counties persists after introducing this detailed set of controls. Column 6 reports an estimated WNS impact of 11.47 basis points and is used to create Figure 1 in the main text.

In Table S4, Panel B, we tabulate the effect of WNS on property tax revenue per capita (columns 1 and 2) and own revenue sources per capita (columns 3 and 4). In these regressions, columns 1 and 3 only include weather controls, while columns 2 and 4 also account for the number of neighboring counties with WNS and the natural logarithm of

population. We use the estimate in column 2 to create Figure 2 in the main text.

2.2. *Heterogeneity.*

In the main text, we explore impacts using important splits of the data, which we discuss later in this section.

2.2.1. *Species richness.*

In Table S5, we explore whether counties that are habitats for a greater number of susceptible bat species experience larger impacts when WNS arrives.

To identify counties with a high number of species at risk of being affected by WNS, we rely on bat species richness data provided by Frank (2024) to create an indicator variable that equals one if a county has an above-median number of bat species susceptible to WNS. The estimates from Table S5 indicate that WNS impacts in counties with more species at risk are larger and more precisely estimated. The estimates presented in Table S5 Panel A and Table S5 Panel B are used in Figure 1, panel (a) and Figure 2, panel (a) in the main text, respectively.

2.2.2. *Property tax assessment frequency.*

In Table S6, we explore whether the impact of WNS varies depending on the frequency of agricultural land value tax assessment updates. Since agricultural land is usually taxed at its agricultural use value (farm revenues minus farm costs), we expect that counties that update their assessments more frequently experience greater revenue changes with the loss of bats to WNS.

We do not observe counties' statutory assessment schedules directly, but we can infer county-level assessment frequency using CoreLogic *Tax* data. First, for each county-year, we calculate the proportion of parcels that show changes in their tax payments compared to the previous year. When this proportion exceeds 50%, we designate that year as a likely assessment year for the county. Second, we calculate the ratio of assessment years to total observed years for each county. Counties with above-average ratios are classified as having high-frequency assessment.

The estimates from Table S6 indicate that counties with more frequent assessments experience larger WNS impacts. The estimates presented in Table S6 Panel A and Table S6 Panel B are used in Figure 1, panel (a) and Figure 2, panel (a) in the main text, respectively.

2.3. *Investment case study results: Bats, local government revenue, and borrowing costs.*

2.3.1. *The investors' opportunities*

In the main text, we illustrate a conservation-focused investor's opportunity to use county municipal bonds as a complement to investing in bat conservation to capture some of the added value of the restored asset value of bats. To simplify the exposition, we assumed that the coupon rate was equal to the average bond yield when WNS was present. This meant that the price of the bond was equal to the face value. However, the data in S1 Panel C show that the average coupon rate is 3.31%. Because $3.04\% < 3.31\%$, using the average coupon rate in equation (SI-3) implies that the bond would be priced at a premium, with a price of $\$1,032,936$ for a $\$1M$ bond. The premium exists because coupon payments exceed

the risk-adjusted opportunity cost of capital. If the spread can be reduced by restoring bats, then the gap between the coupon rate and the yield increases and the premium would increase. Specifically, the price of a county bond in a county without WNS and with an average coupon payment rises to $\$1,047,060$. This change in premium represents a capital gain of $\$14,124$. Overall, a $\$269$ change in capital gains is associated with a 27 basis point change in the coupon rate, suggesting that the qualitative result is not very sensitive to the coupon rate used.

2.3.2. *Municipal governments' conservation opportunities*

Our results suggest that local governments have strong fiscal incentives to invest in biodiversity conservation. Through bat conservation, municipalities can increase their tax revenue by nearly $2.7 million per year on average before considering interest savings.[10] Making conservation investments that maintain higher revenues while lowering borrowing costs means that county governments can continue to provide services, keep taxes low, or both. Conversely, failing to address biodiversity losses will force governments to do some combination of cutting services, raising taxes, or increasing public sector borrowing.

The impact on borrowing costs is also substantial. A municipality facing higher spreads must either increase coupon payments or sell bonds at a discount. These increased costs must ultimately be covered through higher taxes or reduced services. To illustrate the magnitude, in our data, rural counties issue bonds worth an average of $8.34 million (see offering size in Table S1, Panel C). Suppose a municipality that could have issued bonds at

[10] To arrive at $2.7 million, we multiply our estimate of WNS impact on tax revenue from own sources per capita, 149.074 (Table S4, Panel B, column 4), by average rural county population 18,013.22 (Table S1, Panel A).

face value without WNS becomes infected. To continue issuing new bonds at face value, the county would need to raise the coupon payments. For the average county, this would imply an additional $4,784 per coupon payment or $143,537 over 15 years. Multiplying by the 97 rural counties that had experienced WNS as of 2022 suggests national additional annual coupon payments of $928,143, or $27.8 million over 15 years.

Alternatively, if WNS-infected counties paid the average coupon payment prior to infection and wanted to continue issuing bonds at the same price, the average county would need to raise its coupon rate by 11.8 basis points, resulting in an additional $4,917 per coupon payment or $295,021 over 15 years. Across all WNS-infected rural counties as of 2022, this suggests additional annual coupon payments of $953,901, or $28.6 million over 15 years.

2.4. *Supplemental robustness analyses.*

2.4.1. *Methodological robustness.*

In this section, we test robustness to treatment heterogeneity across cohorts, conduct placebo tests with randomly assigned treatment, and examine whether effects are weaker when we repeat our analyses on a sample of urban counties where we expect very limited WNS impact.

First, we examine the robustness of our findings to treatment heterogeneity across cohorts, a concern emphasized in recent econometric literature (see Cunninham (2021) and Roth et al. (2023) for overviews). In Figure S2, we plot regression coefficients from four specifications of our main dependent variables on year dummies relative to WNS

arrival. The figure presents regression results from the two-way fixed effects (TWFE) model with and without a comprehensive set of controls, as well as from two bias-correcting models that account for treatment heterogeneity (Borusyak et al. (2024) and Cengiz et al. (2019)). The Borusyak et al. (2024) approach involves an imputation procedure that models untreated potential outcomes using only untreated observations (never-treated or not-yet-treated) and imputes counterfactuals for treated units, thereby avoiding comparisons to already-treated units. Their estimator is robust to arbitrary treatment-effect heterogeneity and efficient. This approach does not require a balanced panel, allowing us to evaluate the robustness of our bond analyses. An advantage of the stacked design described in Cengiz et al. (2019) is that it treats each cohort as its own sub-experiment and, by construction, uses as controls only units that remain untreated throughout the event window (never-treated or not-yet-treated). Stacking eliminates the TWFE "forbidden comparisons" and negative-weight artifacts.

Figure S2 shows that our estimates are materially unchanged across all approaches, indicating that our results are not driven by weighting or heterogeneous treatment biases that may be present in TWFE.

Second, we perform a placebo test in which WNS treatment and treatment timing are randomly reassigned to counties, and our main specification is re-estimated. This randomization procedure was repeated 10,000 times to generate the distribution displayed in Figure S3. Our estimate from our main specification is marked by the dashed vertical line. The figure indicates that the share of estimates that are greater than 4.423 (2 standard errors below our estimate) is 0.1085. Similarly, the share of estimates above 7.948 (1 standard error below our estimate) is 0.0128. Finally, the share of draws equal to or greater than our

point estimate is 0.0004. This indicates that the probability of us getting our estimate by chance is small.

Finally, to address a concern that our results might be spurious or driven by unobserved economic trends, we repeat our main analyses on a sample of urban counties (defined as non-rural counties; see Figure S1). In Table S7, we find that the effect of WNS on urban counties' yield spreads, property taxes per capita, and own revenue per capita is substantially attenuated toward zero and, for bond yield spreads, is statistically indistinguishable from zero. These null results in urban areas, where the tax base is more diversified and less dependent on agriculture, help validate our main findings.

2.4.2. *Measurement robustness.*

In this section, we assess the sensitivity to choices about the measurement of key variables. We consider a more stringent WNS definition, include nonlinear temperature controls, and measure revenue impacts using different data and scaling approaches.

First, in Table S8, we repeat our main analyses using a more stringent definition of WNS arrival. Specifically, we restrict the "treated" group to counties classified as *WNS Positive* by the White-nose Syndrome Response Team, indicating confirmed presence of the disease by state, federal, or other wildlife management agencies (see section 1.1.1 for a complete description of their classification system). The results are consistent with our main findings.

Second, temperature may impact agricultural outcomes in nonlinear ways (Schlenker and Roberts, 2009). To assess whether any such nonlinearities affect our inferences, in Table S9 we include more flexible temperature controls in our main specifications. In Panel

A, column 1, and Panel B, columns 1 and 3, we include quarterly counts of minimum, maximum, and average daily temperatures in the four temperature bins: $<10\,^{\circ}\mathrm{C}$, $10-20\,^{\circ}\mathrm{C}$, $20-30\,^{\circ}\mathrm{C}$, and $>30\,^{\circ}\mathrm{C}$. In column (2) Panel A, and columns (2) and (4) Panel B, we include linear and squared minimum temperatures to allow for nonlinear temperature effects parametrically. Our main results remain unchanged after including these more flexible temperature controls.

A third potential measurement concern relates to county-level measurement and aggregation of tax data. The measurement of tax data is important because the proposed mechanism behind our main results is that municipal bond investors recognize the financial strain imposed on county finances by the arrival of WNS by demanding higher yield spreads. Because Use-Value Assessment (UVA) systems, which are commonly applied to agricultural land, directly reflect changes in farm profitability, we expect these financial impacts to emerge soon after WNS arrival.

Here, to provide additional evidence supporting this mechanism, we analyze land assessment data from CoreLogic *Tax* to measure the impact of WNS on agricultural tax payments (see also Table S4, Panel B, which establishes a general connection between WNS and property tax collections using Census data).

Table S10 shows that agricultural property taxes paid systematically decrease following WNS arrival in a county. We aggregate observations at the county-year-land use level and estimate our main regression specification using assessed taxes per capita as the dependent variable. In column 1 of Table S10, Panel A, we show that assessed agricultural property taxes per capita decline by \$15.34, a 25% decrease compared to the sample mean. In column 2, we then include our usual controls of the natural logarithm of population and the number

of neighboring counties that have WNS. The results are qualitatively unchanged.

CoreLogic data begin in 2001 and are somewhat sporadically filled until 2016. To assess whether this affects our results, in column 3 of Table S10 Panel A, we repeat our main analyses after limiting the data to begin in 2016. The results remain consistent with our main inferences. In column 4 of Table S10 Panel A, we re-estimate our main specification on the subsample of land uses more likely associated with commercial agriculture (codes above 500). As reported in column 4, the results remain similar but larger in magnitude. This suggests that changes in taxes from commercial agricultural land are a driver of our results.

In Panel B of Table S10, we perform robustness analyses of the results documented in Panel A of Table S10 using alternative dependent variables. In column 1, our dependent variable is the natural logarithm of the assessed taxes on agricultural land in a county. In column 2, our dependent variable is the natural logarithm of the assessed agricultural taxes per acre. In column 3, our dependent variable is the natural logarithm of the number of assessed agricultural acres. Supporting the inferences shown in Panel A of Table S10, we observe consistent negative effects of WNS on each of these three dependent variables. The negative effect on the number of assessed agricultural acres (column 3) is in line with the findings by Manning and Ando (2022), who found that producers decrease the number of agricultural acres in response to the loss of bats to WNS.

A fourth potential measurement concern is how data are normalized by county size. Standard practice is to report financial metrics on a per capita basis, which is our main specification. Here, we verify that our results are robust to alternative normalization methods by rescaling property taxes and own revenues by the number of cropland acres,

using USDA measurements from 1997 (the closest measurement date before the beginning of our sample period). Because this scaling results in a highly skewed variable, we apply a natural logarithm transformation before conducting our regression analyses. The results remain qualitatively consistent with our main findings, demonstrating that our conclusions are not sensitive to the choice of scaling variable (Table S11).

A fifth consideration is that our revenue results may be driven by measurement error. Table S12 extends our revenue analysis in two ways. Panel A provides estimates of the effect of WNS on other county revenue measures, including total tax per capita and total revenue per capita. Both of these variables decline after the arrival of WNS. Panel B presents our main revenue results, but limits the sample of counties to those that issued bonds during the analysis period. The results are qualitatively consistent with our main findings, though slightly larger in magnitude.

A sixth measurement concern relates to the bond spread calculation, which requires specifying a yield curve. The MMA curve is typically preferred as a benchmark because it accounts for the institutional specificities of municipal bonds (Schwert, 2017), but other curves could be used. In Table S13, we present results using spreads calculated over Treasury yields rather than the MMA curve. Our findings remain robust, demonstrating that our main results are not sensitive to the choice of benchmark curve.

2.4.3. Robustness to alternative samples.

In this subsection, we examine whether our results are sensitive to different sample definitions. This includes testing alternative geographic restrictions, more stringent rural county definitions, and samples with different levels of data completeness. These tests

address concerns that our findings might be driven by particular subsets of counties rather than reflecting WNS impacts.

In Table S14, we address the concern that differential trends in the control group might be driving our inferences and re-estimate our main results on a sample that excludes states that never experienced WNS. Our results remain qualitatively unchanged.

All statistical controls used in our main specification (column 6 of Table S4) are estimated on a limited sample because Census population and financial data are fully complete only during mandatory Census years. In other years, some local governments opt not to complete the non-mandatory survey. To assess the role of controls on the magnitude of our estimate, Table S15 reports estimates using a sample that is restricted to observations with non-missing population values. These regressions show that our results are not sensitive to missing population data.

Different analysts may draw different lines between what it means to be a rural or an urban county. Table S16 presents our analysis using a more stringent definition of a rural county. Specifically, we restrict our sample to those counties that have been classified by the USDA as (1) Nonmetro county completely rural or less than 2,500 urban population, not adj. to metro area, (2) Nonmetro county with urban population of 2,500-19,999, not adjacent to a metro area, or (3) Nonmetro county with urban population of 20,000 or more, not adjacent to a metro area. This stricter definition significantly reduces our sample size but still yields results that support our main findings. The larger estimated effects are consistent with our expectation that more rural counties depend more heavily on agriculture.

2.4.4. Robustness to an alternative definition of bats at risk: bat occurrence probability.

Our main analysis follows Frank (2024) and, in one of the heterogeneity analyses, explores variation in impacts by the number of bat species at risk (i.e., susceptible to WNS). Ideally, we would measure declines in bat species that commonly prey on agricultural pests, but those data are not available. We use the richness of susceptible bat species as a proxy because it likely correlates with the number of susceptible individual bats, and pest-controlling species are more likely to be present when more bat species inhabit an area.

One way to confirm this empirically is to correlate the number of species at risk and the probabilities of big brown bat occurrence. The big brown bat is one of the species most affected by WNS, is generally common, a substantial insect predator, and is frequently found over agricultural fields (Fill et al., 2022). We use county averages of detection probabilities from NABat data to test whether increased species richness is associated with higher big brown bat detection probability. We also examine whether counties with greater big brown bat presence experience larger WNS effects.

Big brown bat occurrence probability is correlated with county-level bat richness (Table S17, Panel A). To ensure that our analyses are not confounded by WNS, we restrict this analyses to counties that have never experienced WNS outbreaks. The results demonstrate that big brown bats are more likely to occur in counties with greater numbers of WNS-susceptible bat species and in counties classified as having high bat populations at risk. This supports our use of susceptible bat species richness as a proxy for bats at risk.

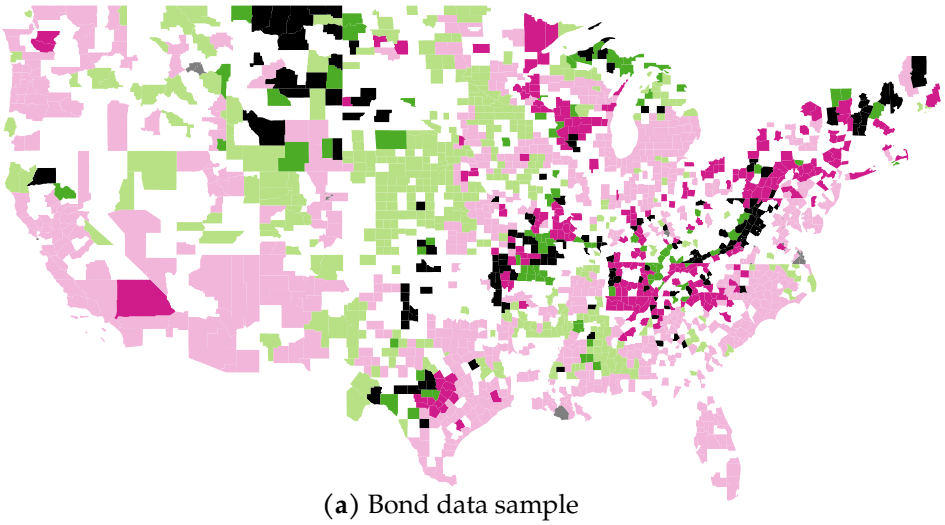
Panels B and C in Table S17 repeat our main specification splitting the sample by median

2016 big brown bat detection probability (2016 being the first year of available occurrence data). To ensure that our results do not capture concurrent WNS trends, we exclude counties that experienced WNS outbreaks after 2016. The findings are similar to our main results reported in Table S5 and Figure 1, panel (a), and Figure 2, panel (a), in the main text.

2.4.5. Robustness to market choice.

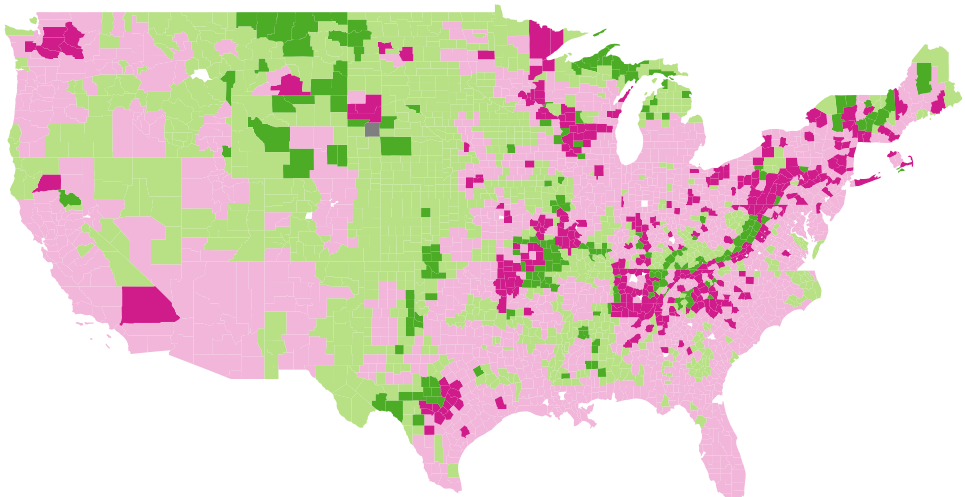
As discussed in the main text, there are data quality and institutional reasons to believe that data from the primary bond market are better suited to our analysis. However, for our investment example to be viable, it must work in the secondary market, where the available data are less well-suited for our analysis. Therefore, we also explore the robustness of our results using secondary market data.

Table S18 presents the results, which are largely consistent with our previous estimates, though the magnitudes are slightly smaller. Column 6 implies that WNS-affected bonds trade at an average premium of 6.7 basis points (an 11% increase relative to the sample median). Unlike in the primary market analysis, we include taxable bonds in this secondary market sample to increase the number of observations. Accordingly, we add a control variable, *Tax Exempt*, which equals one for tax-exempt bonds. Column 7 excludes tax-exempt bonds and confirms that our results are robust to this specification choice.



(**a**) Bond data sample

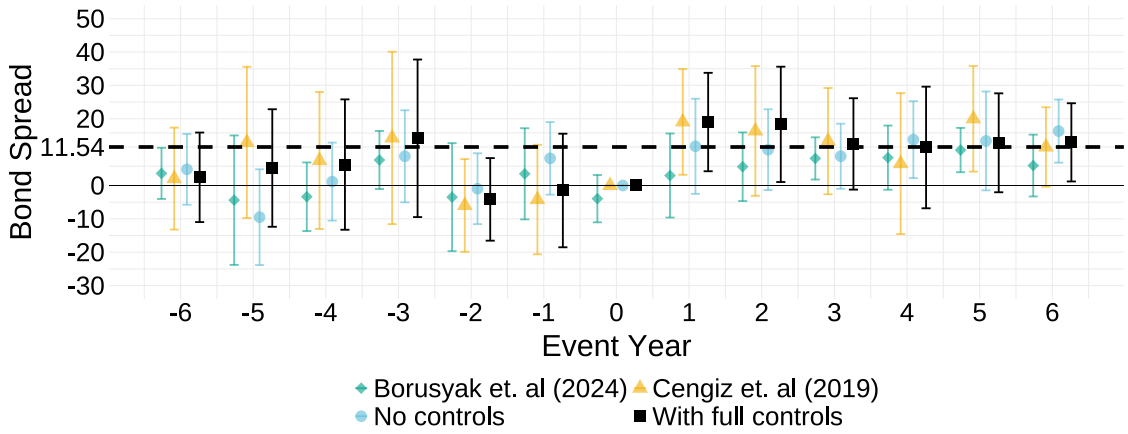
Missing Bond Data no WNS (1337) Rural no WNS (416) Urban no WNS (810)
Missing Bond Data WNS (156) Rural WNS (96) Urban WNS (261)



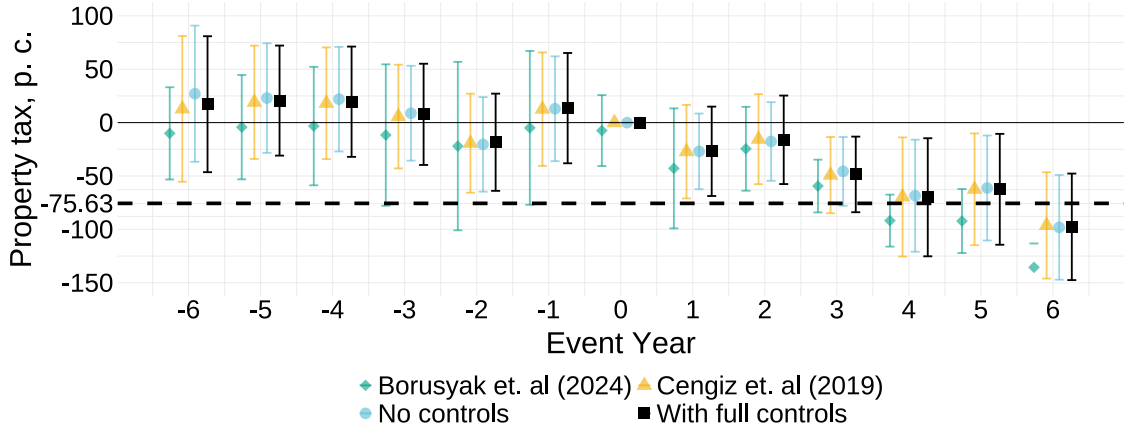
(**b**) Revenue data sample

Missing Fin. Data (91) Rural WNS (195) Urban WNS (329)
Rural no WNS (995) Urban no WNS (1498)

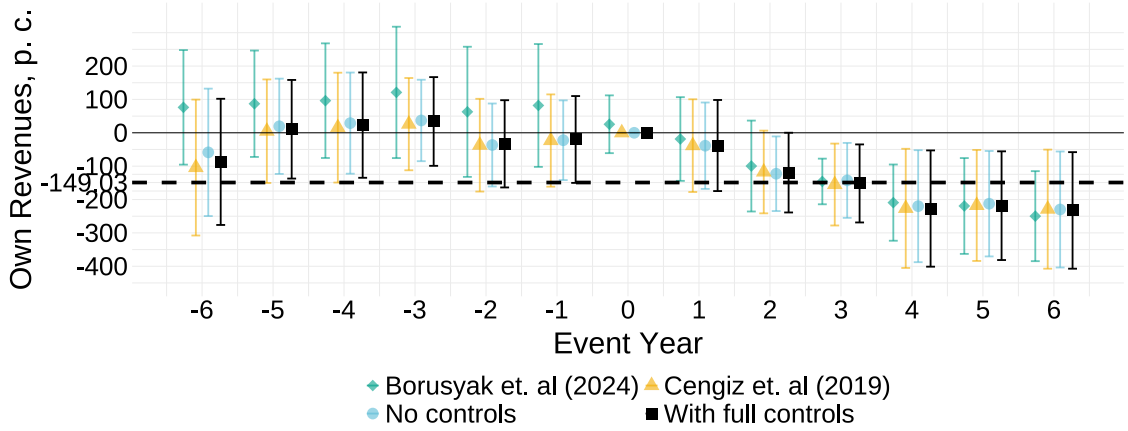
Fig. S1 Distribution of WNS and bond data coverage and tax revenue data coverage across rural and urban counties. This figure displays the map of rural and urban US counties and their respective WNS status. Panel (a) illustrates our bond sample, and panel (b) our tax revenue sample. Rural counties are in shades of green, and urban counties are in shades of pink. Darker shades indicate the presence of WNS in a county as of 2022. Counties with missing data and no WNS are shown in white. Counties with missing data and WNS are in black.



(**a**) Bond spreads



(**b**) Property taxes per capita



(**c**) Own revenues per capita

Fig. S2 Robustness: Sensitivity to heterogeneity across cohorts. Panel (a) of this figure shows bond spreads (in basis points) in event time, Panel (b) displays property tax per capita in event time, and Panel (c) shows own revenues per capita in event time. The baseline two-way fixed effect model without any controls is reported in blue, the model that includes full controls (our main specification) is reported in black, and the two models allowing for treatment heterogeneity across cohorts (Borusyak et al. (2024) and Cengiz et al. (2019)) are reported in green and yellow, respectively. Horizontal dashed lines correspond to our pooled estimates using the full controls specification.

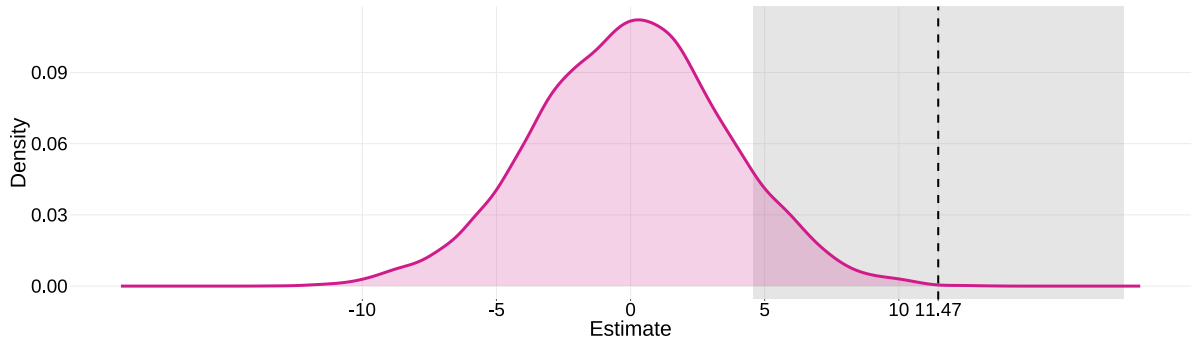


Fig. S3 Robustness: Random WNS assignment. This figure shows the distribution of coefficients from a placebo test in which WNS treatment and treatment timing are randomly reassigned to counties, and the specification from column 6 of Table S4 is re-estimated. This randomization procedure was repeated 10,000 times to generate the distribution. Our actual estimate is marked by the dashed vertical line, and the 95% confidence interval is highlighted by a grey area.

Table S1
Summary statistics for main sample bond characteristics.

Panel A: Descriptive statistics

	Mean	StDev	$p^{10\%}$	$p^{25\%}$	$p^{50\%}$	$p^{75\%}$	$p^{90\%}$	Obs.
Post	0.11	0.31	0.00	0.00	0.00	0.00	1.00	41,268
Spread, bps	42.03	38.89	1.77	17.13	35.36	59.44	88.43	41,268
Bond Size, USD mln	0.55	0.67	0.09	0.17	0.33	0.61	1.23	41,172
Log(Bond Size)	12.68	1.04	11.41	12.04	12.69	13.32	14.03	41,172
Time to Maturity, years	8.85	5.53	2.00	4.00	8.00	12.00	17.00	41,268
Log(Maturity)	1.95	0.74	0.85	1.47	2.07	2.52	2.83	41,268
BQ	0.74	0.44	0.00	0.00	1.00	1.00	1.00	41,268
GO	0.73	0.44	0.00	0.00	1.00	1.00	1.00	41,268
Callable	0.46	0.50	0.00	0.00	0.00	1.00	1.00	41,268
Insured	0.36	0.48	0.00	0.00	0.00	1.00	1.00	41,268
Rating	9.50	8.30	3.00	3.00	5.00	22.00	22.00	41,268

Panel B: Correlations

	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
[1] Post									
[2] Spread, bps	0.050								
[3] Log(Bond Size)	0.130	-0.020							
[4] Log(Maturity)	0.000	0.110	0.220						
[5] BQ	-0.100	-0.040	-0.430	-0.080					
[6] GO	0.090	-0.250	0.000	-0.020	0.020				
[7] Callable	0.010	0.130	0.130	0.720	0.010	-0.050			
[8] Insured	-0.010	-0.160	0.090	0.090	-0.130	0.080	0.010		
[9] Rating	-0.210	0.150	-0.340	-0.230	0.210	-0.060	-0.070	-0.140	

Panel C: Descriptive statistics for hypothetical investment example

	Mean	StDev	$p^{10\%}$	$p^{25\%}$	$p^{50\%}$	$p^{75\%}$	$p^{90\%}$	Obs.
Post	0.11	0.31	0.00	0.00	0.00	0.00	1.00	41,268
Spread, bps	42.03	38.89	1.77	17.13	35.36	59.44	88.43	41,268
Offering Yield, percent	2.75	1.26	1.05	1.70	2.77	3.80	4.35	41,268
MMA Yield, percent	2.50	1.29	0.79	1.42	2.45	3.64	4.21	41,268
Coupon Rate, percent	3.31	1.12	2.00	2.50	3.38	4.00	4.75	41,268
Bond Size, USD mln	0.55	0.67	0.09	0.17	0.33	0.61	1.23	41,172
Time to Maturity, years	8.85	5.53	2.00	4.00	8.00	12.00	17.00	41,268
Offering Size, USD mln	8.34	11.88	1.39	2.60	5.09	9.48	17.98	41,268

Panel A reports summary statistics for the rural county-issued bonds used in the paper. Panel B provides the pairwise Pearson correlations among our bond characteristics of interest. Panel C provides summary statistics for the hypothetical example. *Spread, bps* is the offering yield spread over the MMA AAA curve in basis points. *Bond Size, USD mln* is the bond size (measured in USD million), and *Log(Bond Size)* is its natural logarithm. *Time to Maturity, years* is the bond's time to maturity (measured in years), and *Log(Maturity)* is its natural logarithm. *BQ* is an indicator equal to one if the bond is bank-qualified, which means it has tax-advantaged status when purchased by commercial banks. *GO* is an indicator that equals one for general obligation bonds (bonds backed by the issuer's full taxing authority). *Insured* is an indicator equal to one for insured bonds, where the issuer has purchased a policy, meaning that bondholders will receive repayment of the principal and all interest payments from the insurer if the issuer defaults. *Callable* is an indicator equal to one for bonds with a call option, allowing the issuer to redeem the bond before its maturity date but only after a pre-set call date. *Rating* is the bond's underlying rating across Moody's, S&P, and Fitch (where available), increasing in value from 1 to 22. *Offering Yield, percent* is the bond's yield, in percent. *MMA Yield, percent* is the yield on the MMA AAA curve, in percent. *Coupon Rate, percent* is the coupon rate, in percent. *Offering Size, USD mln* is the total amount borrowed in a single issue (an issue often offers multiple bonds).

Table S2

Summary statistics for main sample county revenue, weather, and population variables.

Panel A: Descriptive statistics

	Mean	StDev	$p^{10\%}$	$p^{25\%}$	$p^{50\%}$	$p^{75\%}$	$p^{90\%}$	Obs.
Post	0.05	0.22	0.00	0.00	0.00	0.00	0.00	12,181
Property tax p.c.	458.80	414.71	86.10	200.07	344.70	566.82	933.58	12,181
Total revenue own p.c.	1,088.90	1,109.84	253.83	400.95	684.73	1,304.93	2,521.69	12,181
Total tax p.c.	568.00	502.28	157.93	262.54	425.05	679.75	1,109.92	12,181
Revenue p.c.	1,602.47	1,475.91	400.50	606.40	1,082.33	2,075.07	3,520.35	12,181
Population	18,013.22	19,219.60	2,529.00	5,289.00	11,253.00	23,206.00	40,828.00	12,181
Log(Population)	9.29	1.06	7.84	8.57	9.33	10.05	10.62	12,181
Neighbors with WNS	0.14	0.43	0.00	0.00	0.00	0.00	1.00	12,181
Q1 avg. min. temp.	-5.20	6.09	-13.01	-9.65	-5.96	-0.44	3.44	12,181
Q2 avg. min. temp.	8.97	4.62	3.09	5.20	8.46	12.84	15.63	12,181
Q3 avg. min. temp.	14.67	4.18	9.00	11.50	14.49	18.26	20.45	12,181
Q4 avg. min. temp.	-0.80	4.97	-6.75	-4.87	-1.88	3.32	6.55	12,181
Q1 avg. precip.	51.82	46.48	9.18	16.12	33.17	78.93	123.44	12,181
Q2 avg. precip.	84.68	45.65	26.64	50.18	80.83	113.63	147.87	12,181
Q3 avg. precip.	75.42	43.86	20.29	42.86	70.66	102.46	135.19	12,181
Q4 avg. precip.	56.77	45.28	12.10	21.19	42.37	82.31	122.25	12,181

Panel B: Correlations

	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	[11]	[12]	[13]	[14]	[15]	[16]
[1] Post																
[2] Property tax p.c.	-0.08															
[3] Total revenue own p.c.	-0.05	0.67														
[4] Total tax p.c.	-0.06	0.95	0.71													
[5] Revenue p.c.	-0.04	0.67	0.90	0.70												
[6] Population	0.13	-0.29	-0.18	-0.24	-0.19											
[7] Log(Population)	0.15	-0.45	-0.26	-0.39	-0.30	0.83										
[8] Neighbors with WNS	0.42	-0.12	-0.08	-0.11	-0.05	0.13	0.12									
[9] Q1 avg. min. temp.	0.02	-0.17	-0.08	-0.11	-0.11	0.21	0.27	-0.02								
[10] Q2 avg. min. temp.	0.02	-0.23	-0.15	-0.20	-0.21	0.17	0.25	-0.00	0.82							
[11] Q3 avg. min. temp.	0.02	-0.22	-0.14	-0.19	-0.20	0.14	0.22	0.01	0.76	0.96						
[12] Q4 avg. min. temp.	0.06	-0.24	-0.13	-0.18	-0.17	0.26	0.33	0.03	0.91	0.90	0.86					
[13] Q1 avg. precip.	0.17	-0.28	-0.18	-0.23	-0.18	0.26	0.36	0.10	0.67	0.61	0.56	0.72				
[14] Q2 avg. precip.	0.18	-0.24	-0.19	-0.24	-0.24	0.14	0.24	0.14	0.31	0.54	0.54	0.46	0.52			
[15] Q3 avg. precip.	0.17	-0.20	-0.15	-0.19	-0.18	0.14	0.23	0.12	0.38	0.59	0.62	0.53	0.51	0.57		
[16] Q4 avg. precip.	0.17	-0.30	-0.19	-0.25	-0.18	0.28	0.38	0.14	0.51	0.53	0.47	0.64	0.77	0.52	0.48	

Panel A reports summary statistics for the revenue, socio-demographic, and weather characteristics of rural counties. Panel B provides the pairwise Pearson correlations of these variables. *Property tax p.c.* is property taxes per capita. *Total revenue own p.c.* is the total revenue from own sources (i.e., excluding transfers from state and federal government), per capita. *Total tax p.c.* is total taxes per capita. *Revenue p.c.* is total revenue (from own sources and from intergovernmental transfers) per capita. *Population* is the population of the county. *Log(Population)* is the natural logarithm of the population of the county. *Neighbors with WNS* is the count of contiguous neighbor counties with WNS. *Q1, Q2, Q3,* and *Q4 avg. min. temp.* are the quarterly averages of daily minimum temperatures measured in Celsius. *Q1, Q2, Q3,* and *Q4 avg. precip.* are the averages of total quarterly precipitation levels in mm.

Table S3

Summary statistics for the sample of urban counties and rural counties.

Panel A: Descriptive statistics for the share of property tax due from agriculture properties as a percentage of all property tax due

	Rural Counties				Urban Counties			
	Mean	StDev	Median	Obs.	Mean	StDev	Median	Obs.
% Property Tax Collected from Agricultural Properties	27.47	25.76	18.81	4,693	11.80	12.12	7.40	8,765

Panel B: Descriptive statistics for bond variables

	Rural Counties				Urban Counties			
	Mean	StDev	Median	Obs.	Mean	StDev	Median	Obs.
Post	0.11	0.31	0.00	41,268	0.09	0.29	0.00	100,027
Spread, bps	42.03	38.89	35.36	41,268	34.80	40.26	26.84	100,027
Bond Size, USD mln	0.55	0.67	0.33	41,172	2.36	3.65	1.02	99,664
Log(Bond Size)	12.68	1.04	12.69	41,172	13.82	1.37	13.84	99,664
Time to Maturity, years	8.85	5.53	8.00	41,268	9.69	5.82	9.00	100,027
Log(Maturity)	1.95	0.74	2.07	41,268	2.04	0.73	2.19	100,027
BQ	0.74	0.44	1.00	41,268	0.19	0.39	0.00	100,027
GO	0.73	0.44	1.00	41,268	0.71	0.45	1.00	100,027
Callable	0.46	0.50	0.00	41,268	0.44	0.50	0.00	100,027
Insured	0.36	0.48	0.00	41,268	0.27	0.44	0.00	100,027
Rating	9.50	8.30	5.00	41,268	5.27	6.84	3.00	100,027

Panel C: Descriptive statistics for financial variables

	Rural Counties				Urban Counties			
	Mean	StDev	Median	Obs.	Mean	StDev	Median	Obs.
Post	0.05	0.22	0.00	12,181	0.08	0.28	0.00	22,891
Property tax p.c.	458.80	414.71	344.70	12,181	341.14	280.18	272.86	22,891
Total revenue own p.c.	1,088.89	1,109.84	684.73	12,181	865.21	777.90	609.12	22,891
Total tax p.c.	568.00	502.28	425.05	12,181	489.94	417.67	373.04	22,891
Revenue p.c.	1,602.47	1,475.91	1,082.33	12,181	1,303.64	1,129.45	900.94	22,891
Population	18,013.22	19,219.60	11,253.00	12,181	191,420.85	486,691.82	64,296.00	22,891
Log(Population)	9.29	1.06	9.33	12,181	11.20	1.26	11.07	22,891
Neighbors with WNS	0.14	0.43	0.00	12,181	0.21	0.59	0.00	22,891
Q1 avg. min. temp.	-5.20	6.09	-5.96	12,181	-1.00	5.88	-1.02	22,891
Q2 avg. min. temp.	8.97	4.61	8.46	12,181	11.88	4.19	11.95	22,891
Q3 avg. min. temp.	14.67	4.18	14.49	12,181	17.15	3.79	17.42	22,891
Q4 avg. min. temp.	-0.80	4.97	-1.88	12,181	3.29	4.70	3.12	22,891
Q1 avg. precip.	51.82	46.48	33.17	12,181	75.38	42.21	71.77	22,891
Q2 avg. precip.	84.68	45.68	80.83	12,181	99.26	48.26	99.50	22,891
Q3 avg. precip.	75.39	43.80	70.66	12,181	97.82	54.61	94.38	22,891
Q4 avg. precip.	56.78	45.33	42.37	12,181	80.78	45.35	76.24	22,891

This table presents summary statistics for rural and urban counties separately. Panel A presents summary statistics for *% Property Tax Collected from Agricultural Properties*, which is a percentage of aggregated agricultural property tax revenues relative to total property tax revenues, computed using CoreLogic data. Panel B reports summary statistics for the bond variables, while Panel C reports summary statistics for the financial, socio-demographic, and weather variables used in the paper. The remaining variables are defined in Tables S1 and S2.

Table S4
Main results.

Panel A: WNS and municipal bond spreads

	Spread					
	(1)	(2)	(3)	(4)	(5)	(6)
Post	10.320***	9.823***	9.514***	9.629***	9.293***	11.471***
	(2.709)	(2.791)	(2.805)	(2.897)	(2.751)	(3.525)
Neighbors with WNS		-0.655	-0.856	-0.793	-0.548	0.913
		(1.704)	(1.674)	(1.759)	(1.779)	(2.356)
Log(Bond Size)			-1.074	-1.878**	-1.873***	-1.851**
			(0.796)	(0.752)	(0.699)	(0.825)
Log(Maturity)			7.295***	6.384***	8.519***	8.915***
			(0.531)	(0.584)	(0.791)	(1.025)
BQ				-15.570***	-14.904***	-17.053***
				(2.340)	(2.083)	(2.487)
GO				-14.031***	-12.003***	-12.782***
				(2.964)	(2.556)	(3.627)
Callable				1.786**	1.175	2.100**
				(0.736)	(0.739)	(0.924)
Insured				-10.276***	-6.680***	-8.153***
				(2.018)	(2.009)	(2.700)
Log Population						-29.773**
						(13.403)
Weather controls	No	Yes	Yes	Yes	Yes	Yes
Mean dependent variable	42.03	42.03	42.03	42.03	42.03	42.03
Number of counties	926	926	924	924	924	697
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Rating fixed effects					Yes	Yes
Adjusted R^2	0.476	0.479	0.496	0.523	0.545	0.551
Observations	41,250	41,250	41,153	41,153	41,152	25,855

Panel B: Revenue analyses

	Property tax p.c.		Total revenue own sources p.c.	
	(1)	(2)	(3)	(4)
Post	-78.136***	-75.638***	-154.020***	-149.074**
	(16.102)	(15.859)	(59.382)	(59.489)
Log Population		-227.912**		-577.114*
		(89.055)		(301.017)
Neighbors with WNS		-3.193		13.759
		(10.473)		(18.096)
Weather controls	Yes	Yes	Yes	Yes
Mean dependent variable	459.05	459.05	1,089.13	1,089.13
Number of counties	1,029	1,029	1,029	1,029
Year fixed effects	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes
Adjusted R^2	0.824	0.825	0.801	0.801
Observations	12,180	12,180	12,180	12,180

This table reports our main results. Panel A presents the results for bond spreads, and Panel B presents the results for revenue variables. Variables are defined in Tables S1 and S2. Standard errors are clustered by county. Levels of significance are presented as follows: *p<0.1; **p<0.05; ***p<0.01.

Table S5
Heterogeneity by bats at risk.

Panel A: WNS and municipal bond spreads

	Spread	
	High	Low
	(1)	(2)
Post	10.470**	7.730
	(4.143)	(8.500)
Neighbors with WNS	0.689	1.614
	(2.598)	(6.216)
Log(Bond Size)	-2.395**	-1.241
	(1.159)	(1.114)
Log(Maturity)	10.147***	8.052***
	(1.340)	(1.461)
BQ	-13.424***	-21.630***
	(3.164)	(3.836)
GO	-15.676**	-12.367***
	(6.967)	(4.092)
Callable	2.840**	1.316
	(1.222)	(1.361)
Log(Population)	-64.683***	-12.809
	(23.328)	(14.202)
Insured	-9.154*	-7.696**
	(4.808)	(3.119)
Weather controls	Yes	Yes
Mean dependent variable	43.31	41.15
Number of counties	318	373
County fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Rating fixed effects	Yes	Yes
Adjusted R^2	0.609	0.513
Observations	12,262	13,369

Panel B: Revenue analyses

	Property tax p.c.		Total revenue own sources p.c.	
	High	Low	High	Low
	(1)	(2)	(3)	(4)
Post	-66.796***	-7.554	-109.709*	-143.193
	(15.563)	(181.786)	(63.429)	(162.224)
Log Population	-117.632	-581.334***	-152.617	-1,456.358***
	(101.681)	(129.943)	(312.445)	(470.283)
Neighbors with WNS	3.995	-53.100	17.614	16.758
	(10.937)	(40.140)	(19.048)	(65.192)
Weather controls	Yes	Yes	Yes	Yes
Mean dependent variable	455.86	476.73	1,061.63	1,172.56
Number of counties	728	291	728	291
Year fixed effects	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes
Adjusted R^2	0.839	0.786	0.816	0.766
Observations	9,038	2,970	9,038	2,970

This table reports analyses by the number of at-risk bat species, with Panel A showing bond spread results and Panel B displaying revenue variable outcomes. Columns labeled *High* show regressions for counties with above-median WNS-susceptible bat species, while *Low* columns report regressions with below-median numbers. Variables are defined in Tables S1 and S2. Standard errors are clustered by county. Levels of significance are presented as follows: *p<0.1; **p<0.05; ***p<0.01.

Table S6
Heterogeneity by assessment frequency.

Panel A: WNS and municipal bond spreads

	Spread	
	High	Low
	(1)	(2)
Post	16.444***	7.180*
	(5.051)	(4.060)
Neighbors with WNS	2.739	0.614
	(3.587)	(2.688)
Log(Bond Size)	-2.804**	-0.613
	(1.297)	(0.853)
Log(Maturity)	9.430***	7.742***
	(1.472)	(0.946)
BQ	-18.904***	-16.239***
	(3.549)	(3.312)
GO	-14.615***	-4.278
	(4.304)	(3.827)
Callable	0.836	4.091***
	(1.215)	(1.363)
Insured	-12.782***	-1.307
	(3.443)	(3.258)
Log(Population)	-27.090	-36.649**
	(17.732)	(16.242)
Weather controls	Yes	Yes
Mean dependent variable	42.75	39.81
Number of counties	452	211
County fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Rating fixed effects	Yes	Yes
Adjusted R^2	0.543	0.582
Observations	15,857	9,165

Panel B: Revenue analyses

	Property tax p.c.		Total revenue own sources p.c.	
	High	Low	High	Low
	(1)	(2)	(3)	(4)
Post	-78.508***	-46.237**	-157.680*	-100.200*
	(19.645)	(21.512)	(87.426)	(54.830)
Log Population	-207.446*	-80.431	-561.724	-218.118
	(116.976)	(93.139)	(398.249)	(434.986)
Neighbors with WNS	-1.189	-15.736	10.967	-15.033
	(13.612)	(14.602)	(19.784)	(34.564)
Weather controls	Yes	Yes	Yes	Yes
Mean dependent variable	444.84	395.58	1,081.12	1,007.41
Number of counties	683	244	683	244
Year fixed effects	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes
Adjusted R^2	0.778	0.883	0.785	0.839
Observations	8,072	2,917	8,072	2,917

This table reports analyses by assessment frequency, with Panel A showing bond spread results and Panel B displaying revenue variable outcomes. Columns labeled *High* show regressions for counties that update assessed values at above-median frequency, while *Low* columns show regressions for counties that update at below-median frequency. Variables are defined in Tables S1 and S2. Standard errors are clustered by county. Levels of significance are presented as follows: *p<0.1; **p<0.05; ***p<0.01.

Table S7
Robustness: Urban counties.

Panel A: Bond spread analyses

	Spread				
	(1)	(2)	(3)	(4)	(5)
Post	0.307	0.858	1.151	1.398	1.452
	(2.838)	(2.825)	(2.810)	(2.875)	(2.723)
Neighbors with WNS		-0.587	-0.606	-0.651	-0.598
		(1.934)	(1.953)	(2.012)	(2.024)
Log(Bond Size)			-0.990**	-1.234***	-1.139***
			(0.440)	(0.389)	(0.351)
Log(Maturity)			12.191***	8.917***	12.160***
			(0.551)	(0.518)	(0.733)
BQ				-10.414***	-9.896***
				(1.602)	(1.562)
GO				-14.202***	-12.069***
				(1.820)	(1.503)
Callable				6.094***	4.722***
				(0.586)	(0.551)
Insured				-8.841***	-7.086***
				(1.693)	(1.689)
Weather controls	No	Yes	Yes	Yes	Yes
Mean dependent variable	34.80	34.80	34.80	34.80	34.80
Number of counties	663	663	663	663	663
County fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Rating fixed effects					Yes
Adjusted R^2	0.428	0.429	0.474	0.497	0.518
Observations	100,018	100,018	99,655	99,655	99,655

Panel B: Revenue analyses

	Property tax p.c.		Total revenue own sources p.c.	
	(1)	(2)	(3)	(4)
Post	-16.415**	-17.662***	-45.081**	-48.561**
	(6.429)	(6.427)	(19.530)	(19.832)
Log Population		-98.428***		-235.573**
		(24.321)		(112.052)
Neighbors with WNS		1.811		12.580**
		(1.766)		(5.955)
Weather controls	Yes	Yes	Yes	Yes
Mean dependent variable	341.14	341.14	865.21	865.21
Number of counties	1,523	1,523	1,523	1,523
Year fixed effects	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes
Adjusted R^2	0.892	0.893	0.867	0.867
Observations	22,890	22,890	22,890	22,890

This table repeats our main results on a sample of urban counties. Panel A presents the results for bond spreads, and Panel B presents the results for revenue variables. Variables are defined in Tables S1 and S2. Standard errors are clustered by county. Levels of significance are presented as follows: *p<0.1; **p<0.05; ***p<0.01.

Table S8
Robustness: Alternative definition of WNS.

Panel A: Bond spread analyses

	Spread					
	(1)	(2)	(3)	(4)	(5)	(6)
Post	8.195***	7.561**	7.437**	6.725**	6.998**	8.967**
	(3.090)	(3.218)	(3.261)	(3.290)	(3.172)	(3.860)
Neighbors with WNS		-0.480	-0.694	-0.608	-0.386	0.998
		(1.714)	(1.681)	(1.755)	(1.769)	(2.297)
Log(Bond Size)			-1.109	-1.913**	-1.904***	-1.885**
			(0.799)	(0.756)	(0.702)	(0.828)
Log(Maturity)			7.316***	6.383***	8.529***	8.932***
			(0.532)	(0.585)	(0.792)	(1.024)
BQ				-15.549***	-14.901***	-17.001***
				(2.348)	(2.090)	(2.477)
GO				-13.889***	-11.870***	-12.626***
				(2.966)	(2.558)	(3.625)
Callable				1.836**	1.215	2.118**
				(0.738)	(0.740)	(0.923)
Insured				-10.272***	-6.679***	-8.220***
				(2.023)	(2.011)	(2.719)
Log Population						-31.988**
						(13.318)
Weather controls	No	Yes	Yes	Yes	Yes	Yes
Mean dependent variable	42.03	42.03	42.03	42.03	42.03	42.03
Number of counties	926	926	924	924	924	697
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Rating fixed effects					Yes	Yes
Adjusted R^2	0.475	0.478	0.495	0.522	0.544	0.549
Observations	41,250	41,250	41,153	41,153	41,152	25,855

Panel B: Revenue analyses

	Property tax p.c.		Total revenue own sources p.c.	
	(1)	(2)	(3)	(4)
Post	-60.836***	-58.427***	-168.446***	-164.354**
	(17.970)	(17.823)	(63.952)	(64.224)
Log Population		-230.194***		-578.973*
		(89.051)		(301.754)
Neighbors with WNS		-3.813		14.920
		(10.527)		(18.376)
Weather controls	Yes	Yes	Yes	Yes
Mean dependent variable	459.05	459.05	1,089.13	1,089.13
Number of counties	1,029	1,029	1,029	1,029
Year fixed effects	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes
Adjusted R^2	0.823	0.824	0.801	0.801
Observations	12,180	12,180	12,180	12,180

This table repeats our main results, restricting the WNS definition to only *WNS Positive* counties, as classified by whitenosesyndrome.org. Panel A presents the results for bond spreads, and Panel B presents the results for revenue variables. Variables are defined in Tables S1 and S2. Standard errors are clustered by county. Levels of significance are presented as follows: *p<0.1; **p<0.05; ***p<0.01.

Table S9
Robustness: Alternative temperature controls.

Panel A: Bond spread analyses

	Spread	
	(1)	(2)
Post	11.934***	11.956***
	(3.567)	(3.622)
Log(Bond Size)	-1.855**	-1.724**
	(0.817)	(0.772)
Log(Maturity)	8.875***	8.907***
	(1.024)	(0.944)
BQ	-16.873***	-16.164***
	(2.443)	(2.530)
GO	-12.638***	-14.132***
	(3.562)	(3.661)
Callable	2.101**	2.097**
	(0.933)	(0.884)
Insured	-7.970***	-8.021***
	(2.674)	(2.842)
Log Population	-31.075**	-24.777*
	(13.078)	(12.945)
Neighbors with WNS	1.050	1.444
	(2.348)	(2.309)
Linear quarterly temp.	No	Yes
Squared quarterly temp.	No	Yes
Quarterly temp. bins	Yes	No
County fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Rating fixed effects	Yes	Yes
Adjusted R^2	0.552	0.565
Observations	25,855	25,855

Panel B: Financial variables

	Property tax p.c.		Total revenue own sources p.c.	
	(1)	(2)	(3)	(4)
Post	-77.641***	-76.223***	-149.696**	-150.986**
	(16.101)	(15.865)	(60.675)	(59.185)
Log Population	-247.310***	-235.523***	-599.469**	-599.181**
	(89.817)	(90.424)	(301.972)	(304.035)
Neighbors with WNS	-2.113	-2.519	13.476	15.281
	(10.497)	(10.476)	(18.349)	(18.150)
Linear quarterly temp.	No	Yes	No	Yes
Squared quarterly temp.	No	Yes	No	Yes
Quarterly temp. bins	Yes	No	Yes	No
Year fixed effects	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes
Adjusted R^2	0.827	0.825	0.802	0.801
Observations	12,180	12,180	12,180	12,180

This table includes alternative temperature controls. Panel A follows the column 6 specification from Table S4 Panel A. Panel B follows columns 2 and 4 specifications from Table S4 Panel B. In column 1 of Panel A and in columns 1 and 3 of Panel B, we include quarterly counts of minimum, maximum, and average daily temperatures in four temperature bins ($<10\,^{\circ}\mathrm{C}$, $10-20\,^{\circ}\mathrm{C}$, $20-30\,^{\circ}\mathrm{C}$, and $>30\,^{\circ}\mathrm{C}$). In column 2 of Panel A and columns 2 and 4 of Panel B, we include squared quarterly average daily minimum temperatures, along with linear ones. Variables are defined in Tables S1 and S2. Standard errors are clustered by county. Levels of significance are presented as follows: *p<0.1; **p<0.05; ***p<0.01.

Table S10
Robustness: WNS and assessed agricultural land taxes.

Panel A: Assessed taxes per capita

	Assessed taxes p.c.			
	(1)	(2)	(3)	(4)
Post	-18.336***	-17.870***	-26.114***	-29.153*
	(3.739)	(3.238)	(3.643)	(15.091)
Log Population		-77.696*	-51.884	-134.892
		(41.229)	(41.374)	(85.866)
Neighbors with WNS		2.160	2.117	4.914
		(1.564)	(1.477)	(3.463)
Weather controls	Yes	Yes	Yes	Yes
Mean dependent variable	70.43	70.43	71.60	140.13
Number of counties	1,052	1,052	1,047	946
Land use more likely ag.	No	No	No	Yes
Post-2016 sample	No	No	Yes	No
Year fixed effects	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes
Land use code fixed effects	Yes	Yes	Yes	Yes
Adjusted R^2	0.579	0.579	0.583	0.669
Observations	18,841	18,841	17,421	7,884

Panel B: Alternative dependent variables

	Log(Assessed taxes)	Log(Assessed tax p.a.)	Log(Acres)
	(1)	(2)	(3)
Post	-0.253***	-0.239***	-0.260***
	(0.055)	(0.054)	(0.048)
Log Population	-0.618	-0.615	-0.397
	(0.716)	(0.706)	(0.660)
Neighbors with WNS	0.011	0.012	-0.019
	(0.033)	(0.032)	(0.039)
Weather controls	Yes	Yes	Yes
Mean dependent variable	10.97	-0.23	7.16
Number of counties	1,052	1,052	1,052
Land use more likely ag.	No	No	No
Post-2016 sample	No	No	No
Year fixed effects	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes
Land use code fixed effects	Yes	Yes	Yes
Adjusted R^2	0.464	0.489	0.604
Observations	18,841	18,841	18,744

This table presents robustness analyses of the impact of WNS on municipal revenues. Panel A presents the assessed taxes per capita as the dependent variable, with alternative sets of controls and fixed effects across the columns. In Panel B, column 1, our dependent variable is the natural logarithm of the absolute assessed taxes on agricultural land in a county. In column 2, our dependent variable is the natural logarithm of the assessed agricultural taxes per acre. In column 3, our dependent variable is the natural logarithm of the number of assessed agricultural acres. The remaining variables are defined in Tables S1 and S2. Standard errors are clustered by county. Levels of significance are presented as follows: *p<0.1; **p<0.05; ***p<0.01.

Table S11
Robustness: WNS and rural county revenues.

	Log(Property tax per cropland)		Log(Total revenue own sources per cropland)	
	(1)	(2)	(3)	(4)
Post	-0.099**	-0.110**	-0.047	-0.057
	(0.049)	(0.049)	(0.043)	(0.042)
Log Population		0.753***		0.734***
		(0.160)		(0.205)
Neighbors with WNS		0.033		0.032**
		(0.021)		(0.016)
Weather controls	Yes	Yes	Yes	Yes
Mean dependent variable	3.41	3.41	4.24	4.24
Number of counties	997	997	997	997
Year fixed effects	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes
Adjusted R^2	0.958	0.958	0.965	0.965
Observations	11,860	11,860	11,859	11,859

This table assesses the sensitivity of our analyses of the impact of WNS on municipal revenues to alternative scaling of dependent variables. The dependent variable in columns 1 and 2, *Log(Property tax per cropland)*, is the natural logarithm of property tax collections, scaled by cropland acres in 1997. The dependent variable in columns 3 and 4, *Total revenue own sources p.c.*, is the natural logarithm of total revenue from own sources per capita, scaled by cropland acres in 1997. The remaining variables are defined in Tables S1 and S2. Standard errors are clustered by county. Levels of significance are presented as follows: *p<0.1; **p<0.05; ***p<0.01.

Table S12
Robustness: WNS and rural county revenues.

Panel A: More aggregate outcome variables

	Total tax p.c.		Revenue p.c.	
	(1)	(2)	(3)	(4)
Post	-77.016***	-74.635***	-214.177***	-215.896***
	(20.397)	(20.235)	(60.583)	(59.921)
Log Population		-187.227*		134.306
		(103.381)		(643.876)
Neighbors with WNS		-7.839		5.792
		(14.157)		(23.356)
Weather controls	Yes	Yes	Yes	Yes
Mean dependent variable	568.41	568.41	1,602.91	1,602.91
Number of counties	1,029	1,029	1,029	1,029
Year fixed effects	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes
Adjusted R^2	0.814	0.815	0.817	0.817
Observations	12,180	12,180	12,180	12,180

Panel B: Restricting sample to counties with bond data

	Property tax p.c.		Total revenue own sources p.c.	
	(1)	(2)	(3)	(4)
Post	-122.901***	-117.025***	-287.595***	-266.499***
	(17.312)	(15.723)	(94.254)	(95.088)
Log Population		-219.273		-707.583
		(138.456)		(520.745)
Neighbors with WNS		13.087		16.185
		(10.039)		(23.158)
Weather controls	Yes	Yes	Yes	Yes
Mean dependent variable	430.00	430.00	1,145.44	1,145.44
Number of counties	512	512	512	512
Year fixed effects	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes
Adjusted R^2	0.831	0.832	0.803	0.804
Observations	6,193	6,193	6,193	6,193

This table presents robustness to the analyses of the impact of WNS on municipal revenues to using more aggregated dependent variables and to restricting the sample to counties with available bond data. Panel A repeats our main results using total taxes per capita, *Total tax p.c.*, (columns 1 and 2) and total revenue per capita, *Total revenue own sources p.c.*, (columns 3 and 4). Panel B repeats our main revenue regressions using a restricted sample that includes only the counties for which we have bond data. The remaining variables are defined in Tables S1 and S2. Standard errors are clustered by county. Levels of significance are presented as follows: *p<0.1; **p<0.05; ***p<0.01.

Table S13
Robustness: Spread over treasury curve.

	Spread over Treasury					
	(1)	(2)	(3)	(4)	(5)	(6)
Post	10.944***	10.790***	10.259**	9.806**	9.467**	12.089**
	(4.086)	(4.102)	(4.051)	(4.116)	(3.945)	(4.967)
Neighbors with WNS		-0.574	-0.774	-0.856	-0.676	0.833
		(2.543)	(2.415)	(2.608)	(2.578)	(3.898)
Log(Bond Size)			0.809	-0.460	-0.419	-0.153
			(0.910)	(0.889)	(0.838)	(0.958)
Log(Maturity)			21.233***	12.888***	15.231***	14.400***
			(0.763)	(0.823)	(1.041)	(1.301)
BQ				-18.221***	-17.464***	-20.879***
				(2.764)	(2.644)	(3.184)
GO				-12.166***	-10.342***	-13.201***
				(3.008)	(2.732)	(3.555)
Callable				16.996***	16.291***	17.416***
				(0.962)	(0.964)	(1.190)
Insured				-11.440***	-7.544***	-10.972***
				(2.659)	(2.596)	(3.444)
Log Population						-37.233**
						(15.718)
Weather controls	No	Yes	Yes	Yes	Yes	Yes
Mean dependent variable	7.73	7.73	7.73	7.73	7.73	7.73
Number of counties	925	925	923	923	923	696
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Rating fixed effects					Yes	Yes
Adjusted R^2	0.606	0.608	0.664	0.682	0.690	0.711
Observations	40,900	40,900	40,804	40,804	40,803	25,629

This table repeats our main bond results, with the dependent variable, *Spread over Treasury*, calculated as the offering yield spread over the Treasury curve and measured in basis points. The remaining variables are defined in Tables S1 and S2. Standard errors are clustered by county. Levels of significance are presented as follows: $^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$.

Table S14
Robustness: Dropping states never affected by WNS.

Panel A: Bond spread analyses

	Spread					
	(1)	(2)	(3)	(4)	(5)	(6)
Post	12.544***	11.731***	11.170***	10.377***	10.312***	13.041***
	(2.886)	(3.051)	(3.063)	(3.102)	(2.931)	(4.113)
Neighbors with WNS		-0.134	-0.289	-0.254	0.169	0.683
		(1.802)	(1.762)	(1.814)	(1.811)	(2.403)
Log(Bond Size)			-0.892	-1.464**	-1.666**	-1.438*
			(0.791)	(0.727)	(0.705)	(0.829)
Log(Maturity)			6.678***	6.048***	7.580***	7.600***
			(0.577)	(0.711)	(0.897)	(1.241)
BQ				-16.597***	-15.036***	-16.439***
				(2.943)	(2.521)	(3.189)
GO				-9.088***	-7.477***	-7.111*
				(3.046)	(2.646)	(3.977)
Callable				1.054	0.854	1.701
				(0.780)	(0.801)	(1.035)
Insured				-11.491***	-7.810***	-7.684***
				(2.157)	(2.185)	(2.860)
Log Population						-12.647
						(22.064)
Weather controls	No	Yes	Yes	Yes	Yes	Yes
Mean dependent variable	42.48	42.48	42.48	42.48	42.48	42.48
Number of counties	614	614	612	612	612	441
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Rating fixed effects					Yes	Yes
Adjusted R^2	0.497	0.499	0.514	0.540	0.562	0.561
Observations	29,254	29,254	29,186	29,186	29,185	16,510

Panel B: Revenue analyses

	Property tax p.c.		Total revenue own sources p.c.	
	(1)	(2)	(3)	(4)
Post	-81.947***	-77.728***	-156.987***	-146.837**
	(16.363)	(15.908)	(60.100)	(59.753)
Log Population		-251.924**		-693.460*
		(112.082)		(384.513)
Neighbors with WNS		0.278		24.370
		(10.137)		(17.104)
Weather controls	Yes	Yes	Yes	Yes
Mean dependent variable	456.01	456.01	1,030.89	1,030.89
Number of counties	819	819	819	819
Year fixed effects	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes
Adjusted R^2	0.824	0.825	0.792	0.794
Observations	9,336	9,336	9,336	9,336

This table repeats our main results on a sample excluding states where WNS was never detected (as of 2022). Panel A presents the results for bond spreads, and Panel B presents the results for revenue variables. Variables are defined in Tables S1 and S2. Standard errors are clustered by county. Levels of significance are presented as follows: *p<0.1; **p<0.05; ***p<0.01.

Table S15
Robustness: Restricting sample to counties with complete population data.

	Spread					
	(1)	(2)	(3)	(4)	(5)	(6)
Post	10.824***	10.264***	10.463***	11.177***	11.744***	11.471***
	(3.722)	(3.758)	(3.794)	(4.018)	(3.602)	(3.525)
Neighbors with WNS		-0.406	-0.650	0.953	0.898	0.913
		(2.196)	(2.158)	(2.428)	(2.359)	(2.356)
Log(Bond Size)			-0.673	-1.713*	-1.908**	-1.851**
			(1.032)	(0.942)	(0.858)	(0.825)
Log(Maturity)			8.005***	6.863***	8.950***	8.915***
			(0.669)	(0.709)	(1.041)	(1.025)
BQ				-17.637***	-17.043***	-17.053***
				(2.899)	(2.496)	(2.487)
GO				-16.394***	-13.073***	-12.782***
				(4.441)	(3.684)	(3.627)
Callable				2.447***	2.072**	2.100**
				(0.936)	(0.933)	(0.924)
Insured				-10.827***	-7.237***	-8.153***
				(2.721)	(2.596)	(2.700)
Log Population						-29.773**
						(13.403)
Weather controls	No	Yes	Yes	Yes	Yes	Yes
Mean dependent variable	40.26	40.26	40.26	40.26	40.26	40.26
Number of counties	698	698	697	697	697	697
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Rating fixed effects					Yes	Yes
Adjusted R^2	0.452	0.460	0.482	0.518	0.549	0.551
Observations	25,894	25,894	25,856	25,856	25,855	25,855

This table repeats our main results on on a sample of rural counties with non-missing population data. Variables are defined in Tables S1 and S2. Standard errors are clustered by county. Levels of significance are presented as follows: *p<0.1; **p<0.05; ***p<0.01.

Table S16
Robustness: Alternative definition of rural county.

Panel A: Bond spread analyses

	Spread					
	(1)	(2)	(3)	(4)	(5)	(6)
Post	14.656**	14.342**	14.074**	13.999**	13.900**	16.012*
	(6.078)	(6.177)	(6.294)	(6.374)	(6.577)	(8.319)
Neighbors with WNS		1.351	0.912	0.655	0.202	-2.746
		(3.939)	(3.900)	(4.000)	(3.986)	(5.294)
Log(Bond Size)			-2.928**	-3.150**	-2.779**	-2.993**
			(1.487)	(1.343)	(1.160)	(1.317)
Log(Maturity)			7.668***	7.700***	9.895***	10.021***
			(0.858)	(0.900)	(1.031)	(1.340)
BQ				-17.327***	-16.711***	-18.018***
				(4.119)	(3.179)	(3.559)
GO				-11.017***	-8.779***	-8.341**
				(3.662)	(2.801)	(4.128)
Callable				-0.082	-0.522	0.423
				(1.150)	(1.014)	(1.397)
Insured				-14.832***	-10.796***	-13.385***
				(3.389)	(3.431)	(4.908)
Log Population						-37.143*
						(18.965)
Weather controls	No	Yes	Yes	Yes	Yes	Yes
Mean dependent variable	45.59	45.59	45.59	45.59	45.59	45.59
Number of counties	432	432	430	430	430	312
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Rating fixed effects					Yes	Yes
Adjusted R^2	0.479	0.487	0.503	0.529	0.565	0.583
Observations	17,409	17,409	17,352	17,352	17,351	10,500

Panel B: Revenue analyses

	Property tax p.c.		Total revenue own sources p.c.	
	(1)	(2)	(3)	(4)
Post	-86.820***	-84.513***	-154.366**	-150.086**
	(18.439)	(18.163)	(73.277)	(73.578)
Log Population		-196.561**		-513.787
		(98.411)		(334.410)
Neighbors with WNS		-3.964		13.273
		(11.586)		(20.642)
Weather controls	Yes	Yes	Yes	Yes
Mean dependent variable	458.49	458.49	1,125.34	1,125.34
Number of counties	831	831	831	831
Year fixed effects	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes
Adjusted R^2	0.823	0.824	0.800	0.801
Observations	10,171	10,171	10,171	10,171

This table replicates our main results using rural counties under a more stringent rural classification (defined in section 2.4). Panel A presents the results for bond spreads, and Panel B presents the results for revenue variables. Variables are defined in Tables S1 and S2. Standard errors are clustered by county. Levels of significance are presented as follows: *p<0.1; **p<0.05; ***p<0.01.

Table S17
Robustness: Bat occurrence probability and main outcomes.

Panel A: Big brown bat and richness

	Big Brown Bat Occurence	
	(1)	(2)
N Species Susceptible to WNS	0.020***	
	(0.002)	
High Bats at Risk		0.026***
		(0.006)
Constant	0.437***	0.499***
	(0.008)	(0.004)
Adjusted R^2	0.046	0.007
Observations	2,507	2,507

Panel B: Bond spreads

	Spread	
	High	Low
	(1)	(2)
Post	16.319***	5.286
	(4.333)	(7.039)
Neighbors with WNS	-0.304	0.918
	(2.481)	(5.986)
Log(Bond Size)	-1.988**	-2.414*
	(0.822)	(1.403)
Log(Maturity)	9.520***	9.627***
	(1.090)	(1.269)
BQ	-19.614***	-18.215***
	(3.067)	(3.737)
GO	-10.013**	-16.591***
	(4.096)	(4.848)
Callable	1.989*	1.470
	(1.029)	(1.327)
Insured	-5.286*	-11.052**
	(2.922)	(4.412)
Weather controls	Yes	Yes
Mean dependent variable	40.60	40.59
Number of counties	311	343
County fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Rating fixed effects	Yes	Yes
Adjusted R^2	0.592	0.542
Observations	12,359	12,547

This table shows the correlation between big brown bat occurrence and the number of bat species susceptible to WNS (Panel A). Panels B and C repeat our main results on species richness, splitting by whether the 2016 big brown bat occurrence probability was high (above median) or low (below median). We restrict the treatment definition to counties that were exposed to WNS prior to 2016. Variables are defined in Tables S1 and S2. Standard errors are clustered by county. Levels of significance are presented as follows: *p<0.1; **p<0.05; ***p<0.01.

Table S17
Robustness: Bat occurrence probability and main outcomes, *continued.*

Panel C: Revenue analyses

	Property tax p.c.		Total revenue own sources p.c.	
	High	Low	High	Low
	(1)	(2)	(3)	(4)
Post	-112.728***	-73.911***	-182.931**	-165.747***
	(14.111)	(22.024)	(90.908)	(43.020)
Log Population	-279.985*	-296.858***	-654.926*	-1,067.543***
	(143.848)	(74.146)	(362.657)	(296.091)
Neighbors with WNS	-3.308	-11.066	20.425	-12.715
	(13.094)	(12.863)	(22.189)	(27.822)
Weather controls	Yes	Yes	Yes	Yes
Mean dependent variable	455.43	442.99	1,018.38	1,133.23
Number of counties	448	493	448	493
Year fixed effects	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes
Adjusted R^2	0.818	0.843	0.812	0.806
Observations	5,613	5,817	5,613	5,817

This table shows the correlation between big brown bat occurrence and the number of bat species susceptible to WNS (Panel A). Panels B and C repeat our main results on species richness, splitting by whether the 2016 big brown bat occurrence probability was high (above median) or low (below median). We restrict the treatment definition to counties that were exposed to WNS prior to 2016. Variables are defined in Tables S1 and S2. Standard errors are clustered by county. Levels of significance are presented as follows: $^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$.

Table S18
Robustness: Secondary market analysis.

	Spread						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Post	6.353*	6.211*	5.875	6.880**	6.690**	6.679**	6.321*
	(3.685)	(3.676)	(3.685)	(3.306)	(3.340)	(3.382)	(3.249)
Neighbors with WNS		-0.257	-0.184	-0.140	-0.045	-0.045	0.309
		(1.431)	(1.428)	(1.398)	(1.408)	(1.406)	(1.516)
Log(Maturity)			2.107	-4.323***	-4.101***	-4.101***	-4.014***
			(1.339)	(1.534)	(1.505)	(1.507)	(1.491)
BQ				-13.785***	-13.760***	-13.759***	-13.597***
				(2.004)	(1.973)	(1.973)	(1.842)
GO				1.864	2.304	2.305	-0.513
				(5.064)	(4.963)	(4.961)	(4.396)
Callable				8.768***	8.596***	8.596***	9.840***
				(2.007)	(1.989)	(1.989)	(1.895)
Tax Exempt				-80.956***	-80.980***	-80.977***	
				(5.332)	(5.362)	(5.368)	
Insured					-4.725**	-4.728**	-5.748***
					(2.295)	(2.313)	(2.101)
Log Population						-0.726	-16.450
						(28.047)	(29.311)
Tax-exempt sample	No	No	No	No	No	No	Yes
Weather controls	No	Yes	Yes	Yes	Yes	Yes	Yes
Mean dependent variable	73.33	73.33	73.33	73.33	73.33	73.33	67.73
Number of counties	477	477	477	477	477	477	453
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Trade Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Rating fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.375	0.375	0.376	0.450	0.450	0.450	0.325
Observations	32,138	32,138	32,138	32,138	32,138	32,138	30,243

This table repeats our main bond results in the secondary market setting. *Tax Exempt* is an indicator equal to one if the bond is tax exempt. The remaining variables are defined in Tables S1 and S2. Standard errors are clustered by county. Levels of significance are presented as follows: *p<0.1; **p<0.05; ***p<0.01.

References

Borusyak, K., Jaravel, X., Spiess, J., 2024. Revisiting event-study designs: Robust and efficient estimation. The Review of Economic Studies 91, 3253–3285.

Cengiz, D., Dube, A., Lindner, A., Zipperer, B., 2019. The effect of minimum wages on low-wage jobs. The Quarterly Journal of Economics 134, 1405–1454.

Cestau, D., Hollifield, B., Li, D., Schürhoff, N., 2019. Municipal bond markets. Annual Review of Financial Economics 11, 65–84.

Cornaggia, J., Cornaggia, K. J., Israelsen, R. D., 2018. Credit Ratings and the Cost of Municipal Financing. Review of Financial Studies 31, 2038–2079.

Cunninham, S., 2021. Causal inference: the mixtape. Yale University Press.

Dagostino, R., Nakhmurina, A., 2023. Risk sharing in a political union. Available at SSRN 4480061 .

Fill, C. T., Allen, C. R., Twidwell, D., Benson, J. F., 2022. Spatial distribution of bat activity in agricultural fields: implications for ecosystem service estimates. Ecology and Society 27.

Frank, E. G., 2024. The economic impacts of ecosystem disruptions: Costs from substituting biological pest control. Science 385.

Green, R. C., Li, D., Schürhoff, N., 2010. Price discovery in illiquid markets: Do financial asset prices rise faster than they fall? The Journal of Finance 65, 1669–1702.

Manning, D. T., Ando, A., 2022. Ecosystem services and land rental markets: producer costs

of bat population crashes. Journal of the Association of Environmental and Resource Economists 9, 1235–1277.

Painter, M., 2020. An inconvenient cost: The effects of climate change on municipal bonds. Journal of Financial Economics 135, 468–482.

Pierson, K., Hand, M. L., Thompson, F., 2015. The government finance database: A common resource for quantitative research in public financial analysis. PloS one 10, e0130119.

Roth, J., Sant'Anna, P. H., Bilinski, A., Poe, J., 2023. What's trending in difference-in-differences? a synthesis of the recent econometrics literature. Journal of Econometrics 235, 2218–2244.

Schlenker, W., Roberts, M. J., 2009. Nonlinear temperature effects indicate severe damages to us crop yields under climate change. Proceedings of the National Academy of sciences 106, 15594–15598.

Schwert, M., 2017. Municipal Bond Liquidity and Default Risk. Journal of Finance 72, 1683–1722.

Udell, B. J., Straw, B., Cheng, T. L., Enns, K., Frick, W. F., Gotthold, B., Irvine, K., Lausen, C., Loeb, S., Reichard, J. D., et al., 2022. Status and trends of north american bats: Summer occupancy analysis 2010-2019. USGS Report p. 40.